

Software Engineering

04GSPOV

Course Information

What

- What will I learn in this course?

The software 'factory'



software

Why

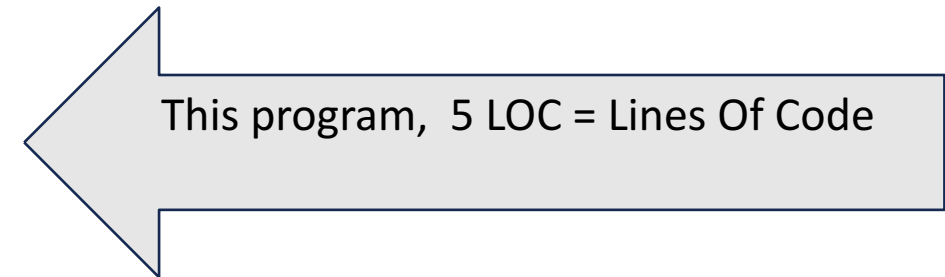
I can write a {C, Java, Python, Javascript, PHP,} program, what else should I need to know?

Software engineering is about software as a **commercial, industrial product**

What was the largest program you coded?

taking user input

```
ch = ("Enter a character: ")
if ((ch>='a' and ch<= 'z') or (ch>='A' and ch<='Z')):
    (ch, "is an Alphabet")
else:
    (ch, "is not an Alphabet")
```

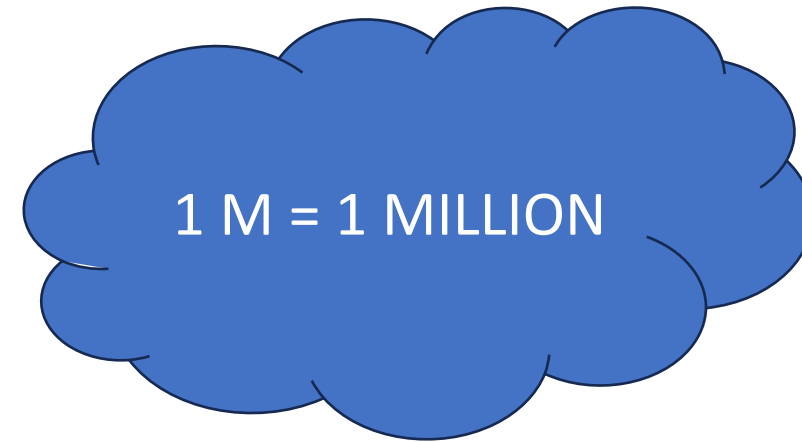


- www.menti.com
- Code **7984 3526**



Size of 'real' programs

- Linux OS: > 15 M LOC
- Android OS: > 12 M LOC
- Windows : > 50 M LOC
- Boeing 787: > 7 M LOC
- Car (high level models): > 100M LOC
- ERP : > 1500 RDB tables



Why

Software as an industrial product

- Is large, and requires a **team**, not a single programmer
 - need to share programs and documents
 - configuration management, repository tools
 - need to manage the team
 - project management, team management

Why

Software as an industrial product

- Is made for a **customer**, who has **no notion of computing**
 - need to understand what the customer wants / needs
 - requirement engineering
 - need to define and respect deadlines and contracts
 - project management
 - need to assure a defined level of quality
 - testing

Why

Software as an industrial product

- Is usually a **large and complex** product
 - need to master the complexity, and implement what was promised to the customer
 - software design
 - project management

Why

Software as an industrial product

- Is an asset with a **great financial value** that must last years
need to maintain and evolve the product for years
 - software maintenance
 - software design
 - testing
 - configuration management

Why

Software as an industrial product

- Is an asset with **critical effect on the customer's business**

need to avoid critical defects

→ software testing

→ software quality

Critical failures

- British airways (2017)
 - Power surge failure
 - Back up system fails to engage
 - Information systems down at Gatwick and Heathrow
 - 1000 flights canceled over 3 days, 75000 people affected
 - 200M\$ compensation costs
 - British Airways stock exchange loss 200M\$

Critical failures

- ArianeV rocket launch failure, June 1996
- Due to a system / software defect



Why

Software as an industrial product

- Requires a large number of activities and tools, that must be coordinated and organized effectively
→ software process



Why

- All right, developing software is a f...g mess, and that is why i dont want to do it in my career

Why

Software Is Eating the World



Why

- Software is everywhere!



Why

- Every person should be computer literate
 - Reading, writing, math, COMPUTING
 - Digital divide
 - Social and work exclusion of citizens who are computer illiterate
- Every engineer should be (way more than) computer literate

Why

- Every company should become a software company
 - Amazon vs. Border's
 - Netflix vs. Blockbuster
 - Apple, Google vs. Toyota, Volkswagen, GM, ..

Why

- The economies of ALL developed nations are dependent on software.
- Expenditure on software represents a significant fraction of GNP in all developed countries.
- There is a clear relationship between innovation, productivity, wealth and expenditure in IT technology

Who

- Maurizio Morisio
 - Dept. Control and Computer Engineering
 - maurizio.morisio@polito.it
- Tommaso Fulcini
 - Tommaso.Fulcini@polito.it

When (and where)

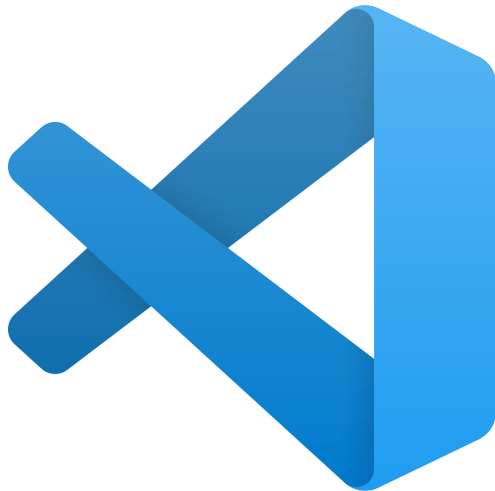
Day	Room
Monday 8.30	R4
Wednesday 8.30	R4
Thursday 11.30	R2
Friday 11.30	R3

- BYOD (Bring Your Own Device)!
 - Classrooms have plugs
- All the lectures will be streamed and recorded

How

- Lectures and Exercises
 - In classroom, BYOD
- Participation is important
 - live / in streaming / on demand

How



GitLab



CATME
SMARTER Teamwork



What will I learn?

- Software Engineering
 - Motivation, issues, problems
- Requirement engineering
 - How to understand what the customer / user needs
- Design
 - How to put together components which deliver what the customer needs
- Verification and Validation
 - How to assure that the product does what it is supposed to do
- Project management
 - How to manage teamwork and deliver the product
- Software process
 - How to organize activities and tools, what process models are available

How to... pass the exam

- Two parts
 - Project work (explained in detail later)
 - Exam
- Final grade (max 33 points) =
grade of project (max 15 points)
+
grade written exam (max 20 points).

The project grade $\geq 12 / 20$

In case the project is not done, max possible grade is 20/30
30cum laude requires ≥ 32.5 points

Books and References

- Bruegge Dutoit, Object Oriented Software Engineering
- Ian Sommerville, Software Engineering, Pearson ed.
 - Also, Pfleeger, Pressman, Shach
- Martin Fowler, UML Distilled

And then?

- How does Software development work in practice?
 - Modern teams organize with **agile** practices
- Quality is a key component in engineering practice, how does it work in Software engineering?
 - Software **analytics** allow measuring and controlling Sw projects
- Evolution is intrinsic in software, how do you tame it?
 - Advanced **debugging** methods, **log** analysis, **reverse engineering** techniques

Software Engineering 2!

- Professors: Marco Torchiano, Antonio Vetrò
- A practical, project-oriented course to learn how software is crafted in modern teams
- Let's try in practice
 - Software Scrum
 - Planning poker
 - Technical debt
 - Software smells

Project

Process and product

- Apply a state-of-the-art software engineering process to a (small) project:
 - Use of tools
 - Use of techniques
 - Use of process
 - Work in a team
- Work on a project, code, document, test and modify it

Constraints

- Project must be developed in parallel with course
- The object to be developed is the same for all the teams
 - Discussion inside the team is essential
 - Copying between teams is forbidden
- Antiplagiarism tools will be applied
- All communications only via defined tools
 - (mostly the Git repository, or Slack)

Steps

1. Teams will be defined by us
 - You cannot choose your teammates (as it happens in real life!)
2. Set up repository
 - You will receive an email (@polito account) for setting up your account and team repository
3. Access objects of development on repository
4. Produce various deliverables

To be produced

- Requirements document
- Design document
- Code
- Test suite – unit level, integration level
- Test suite, API level
- Change Request